A DTD for Qualitative Data: Extending the DDI to Mark-up the Content of Non-numeric Data

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Why another DTD?

- need a standard
  - that includes both file-level metadata and content-level metadata
    - enables more precise searching/browsing
    - extends to linking between sources (e.g. text, annotations, analysis, audio etc)
- need one customised to social science research that:
  - meets generic needs of varied data types
  - is more 'analytical' than ones adapted from TEI speech schema (e.g. oral history projects)
  - is less granular than ones for conversational analysis (highly detailed)

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Specific applications

- marking up data to an XML standard for data providers to publish to online systems, such as ESDS Qualidata Online (formerly Edwardians)
- meet needs of researchers requesting a standard they can follow
- encourage more qualitative data analysis software companies to pursue XML- outputs (and import/export tools) based on this standard

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Hybrid of two standards

for the metadata – the DDI Standard for study, file and variable level

- Level 1: DDI Document description
- Level 2: DDI Study description
- Level 3: DDI Data file description
  - file contents; format; data checks; processing; software)
- Level 4: DDI Variable description:
  - for study survey data (mixed methods) or numeric outputs from qualitative data:
    - demographic profile of sample
    - other quantified responses to qualitative data (attributes or thematic classifications often assigned (coded) in CAQDAS software)
- Level 5: DDI Other Study related materials
- Level 6: TEI-based qualitative content

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TEI for content mark-up

- standard for text mark-up in humanities and social sciences

- Elements for the header for a TEI-conformant DTD:
  <teiheader = type = text/corpus>
  <fileDesc> <encodingDesc> <profileDesc>
  <revisionDesc> standard bibliographic ref to text
- Mandatory =

```
<teiHeader type=text>
  <fileDesc>
        <titleStmt> <!-- ... --> </titleStmt>
        <publicationStmt><!-- ... -->
        </publicationStmt>
        <sourceDesc> <!-- ... --> </sourceDesc>
  </fileDesc>
  <!-- remainder of TEI Header here -->
</teiHeader>
```

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Four components of a TEI DTD

- core tag set – available to all TEI docs
- base tag set – Transcription of speech

  <!ENTITY % TEI.spoken 'INCLUDE' >
- additional tag sets – optional
  - linking
  - analysis
  - certainty and responsibility
  - transcription
  - names and dates
  - corpora
- entity tag sets – not needed

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Issues this DTD resolves

- multiple speakers
- turn taking
- researcher annotations of transcripts
- thematic coding (as well as is possible with XML)
- name and place references
- compatibility with existing XML-enabled qualitative data analysis software (e.g. Atlas.ti output)
- As always, formatting elements handled with style sheets, not in the DTD

Much work remains…

- Further integration of DDI and TEI required elements
- Define the DTD for an individual case (e.g. transcript) or a collection, or both?
- Elements selected: not too many, not too few – assign mandatory and optional
- How elements are used: follow existing norms, set standard where necessary

✓ Need DDI specialist interest group/DDI structural reform group to help define and refine a suitable DTD

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Proposed elements and samples

- See Table of Proposed Elements
- Sample case-level XML (transcript) marked up with a subset of proposed elements
- Sample study-level XML using DDI standard (levels 1-3 and 5)
- Draft DTD soon available on ESDS Qualidata website

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Excerpt from interview transcript

My father was, in the daytime he was a boiler maker on the old North ~~Circular~~ Staffordshire Railway and then every night he played in the theatre orchestra. And sometimes even after the theatre he would go on andplay for an hour or two at a dance, well they called them balls in those days. And he ~~had got to be at~~ a to go to work at six the next morning! Cornet player.

Goodness. Would you see him when he used to play?

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Excerpt with XML mark-up

```
<u n="31">
…
<s n="44">  My father was, in the daytime he was a boilermaker
    on the old
    <name type="organisation">
    North <add place="supralinear">Staffordshire</add>
    <del type="word change">Circular</del>
    Railway</name>
    and then every night he played in the theatre orchestra.
</s>

<s n="45">  And sometimes
    <add place="supralinear">even</add>
    after the theatre he would go on and play for an hour or two
    at a dance, well they called them balls in those days.
</s>

<s n="46">And he
    <add place="supralinear">'d to go to</add>
    <del>had got to be at</del> work at six the next morning!
    <note place="end of paragraph">Cornet player.</note>
</s>
</u>
```

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Thematic coding: Stand-off Architecture in XML

- Challenges for developing an XML application included the multiple hierarchies in the transcript texts and overlapping fields or elements:

  dialogue structure v thematic content

- Conventional mark-up of these structures in a single document violates nesting rules of XML
- Solution - 'stand-off annotation' approach whereby data and coding stored in different documents (annotation linked by Xlink and Xpointers)
- Proven utility as method for annotating multi-coded dialogue corpora. Allows for:
  - multiple coding schemes
  - overlapping elements
  - easily extendable

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Base-line text unit: utterances (<u>)

<u> attributes:

- id
- speaker …
- start time (audio file)
- end time (audio file)

157 Did your father take an interest in politics ?

158 Yes, my father was in those days known as a Radical. And he was a real right Radical too, I'll tell you, yes. 'Cos my father lost his job at - Tillings through a - the men - through - through sticking up so long - so much for his men, getting with the unions, yes, he got - yes, he lost his job. When they discharged the men they discharged him as well. Oh he was a real right Labour, well later years he was a Labour - Labour man. Oh he was - was a shocking heckler. Oh. Terrible.

159 Do you remember if he ever told you about his views or why he held them, do you know what made him such a Radical?

160 No, I don't know really 'cos he'd had rather a good bringing up, he'd been at college and all that sort of thing, but of course his parents weren't badly off, but - I don't know but he was a real - it was - because he used to belong to what they called a Radical club, and he was one of the officials there, I mean, he - he was really - up in it. Yes. Yes. Yes.

161 Do you know whether he was a member of a party?

162 Well I don't know - I don't know whether you would call it a member of the party or not, but I know that he was always - when they went - 'cos - before Labour came in 'cos he was always like liberal, 'cos there was no Labour then, but this Radical business was coming on a long time before that, and I can remember 'cos we used to - we used to go to Christmas parties that were run by the Radical people you see.

163 He used to work for the liberals at election time?

164 Yes, I suppose - yes. Oh yes, we always had to wear -

Theme: work

Theme: household

Theme: politics

Example of 'Stand-off' XML Architecture

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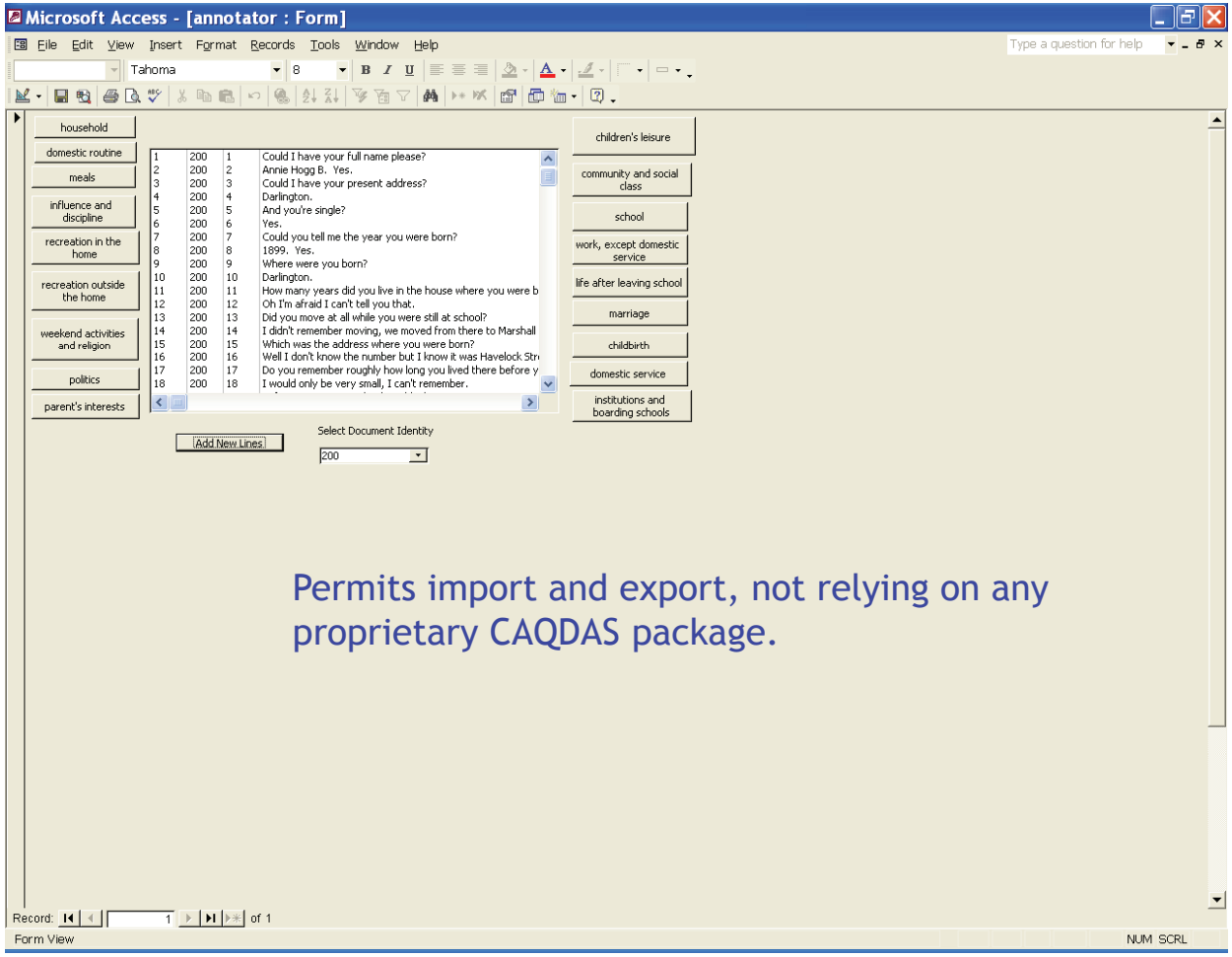
line_id	doc_id	u_id	who	utterance
171	200	173	inter	Did anyone else besides your parents and brothers and sisters live in the house, did you have any relatives or lodgers living with you?
172	200	174	subj	We had my grandma came.
173	200	175	inter	How old would you be when your grandmother came to live with you?
174	200	176	subj	Well I should only - think I would only be about five year old.
175	200	177	inter	How long did your grandmother live with you?
176	200	178	subj	Oh dear - she - until our Margaret was born and she was - the last one.
177	200	179	inter	So she lived with you quite a long time?
178	200	180	subj	Yes, she did. Oh Margaret's - I can tell you Margaret's age. She's sixty-one.
179	200	181	inter	So that was 1905 - 1909?

line_id	theme_id
171	2
171	11
172	2
172	11
173	2
174	1
174	2
175	1
175	2
176	1
176	2
177	1
177	2
178	1
178	2
179	1
179	2

Coding of utterance (turn of speech) by theme

In-house tool for coding themes



Permits import and export, not relying on any proprietary CAQDAS package.

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Selected elements from Atlas for codes (themes) and pointers

```
<codes size="52">
<code
    name="A Formula"
    id="co_5"
    au="Thomas M"
    cDate="2003-03-04T14:30:57"
    mDate="2003-03-07T13:19:42"
    cCount="0"
    qCount="1" >
</code>

<q
name="And the name of the star is ca..“
id="q1_1"
au="Admin"
cDate="1991-03-11T13:27:48“
mDate="1993-10-08T21:45:00"
loc="5 @ 27, 98 @ 27"/>
</q>
```

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What does the DTD enable?

- ability for data producers to publish data in multiple formats using style sheets/using web-based systems
- e.g. ESDS Qualidata Online – brief demo
  http://www.esds.ac.uk/qualidata/online/explore/transcriptsmultiple.asp
- enable data exchange and data sharing across dispersed repositories (c.f. Nesstar)
- Enable the development of import/export functionality for CAQDAS software

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Need for publishing tools

- Once DTD is more devloped, next step is to develop publishing tools to automate as much of mark-up as possible
- Currently using simple scripts to find and mark <u> and <s>; much work still done manually
- Looking into options for automatic mark-up of some components (e.g. natural language processing and information extraction):
  - Brill tagger
  - Gate architecture http://gate.ac.uk
  - Customising existing NLP tools at Sheffield and Edinburgh

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Collaborators

- Oxford Computer Centre (TEI)
- NLP team at Sheffield
- NLP team at Essex
- NLP team at Edinburgh
- Atlas.ti developers (Berlin)
- Cardiff Ethnography Group
- E-social science programme text mining groups
- Academics in UK who wish to use standard
- FSD
- US and rest of world?
- DDI, IASSIST, CESSDA

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Selected References

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For more information

- ESDS Qualidata

  http://www.esds.ac.uk/qualidata/introduction.asp

- ESDS Qualidata Online

  http://www.esds.ac.uk/qualidata/online/

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